Gulsher Assad

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EDUCATION

The University of British Columbia — Faculty of Science Kelowna, BC
Bachelor of Science (BSc), Computer Science *Graduation Expected August 2027*
Relevant Coursework: Data Structures & Algorithms, Machine Architecture, Software Engineering, Statistical Modelling, Network Science, Introduction to Databases, Learning Analytics

PROJECTS

Scout - Football Player Similarity Engine | *Python, FastAPI, scikit-learn, pandas, TypeScript, React, pytest* September 2026

- Developed a football scouting tool that identified statistically similar attackers across Europe's top 5 leagues, profiling 522 players from 44,000+ shots using per-90 statistics, chance-creation metrics, and shot-location features with NMF.
- Improved similarity recall@10 from 0.205 to 0.258, achieving over 12x the random baseline by designing a split-half evaluation benchmark across 20 test splits.
- Engineered an incremental data pipeline with validation checks that detected incorrectly recorded own goals and duplicate matches, then linked players to Transfermarkt market values using fuzzy name matching.

Task Manager API | Java · Spring Boot · Maven · Spring Data JPA · Hibernate · H2 · JUnit · REST August 2026

- Built a RESTful task-management API in Java and Spring Boot with full CRUD, status/priority filtering, and due-date sorting over a layered controller-service-repository architecture.
- Enforced input integrity with Bean Validation that rejects malformed requests at the API boundary, backed by database column constraints.
- Wrote 6 JUnit and MockMvc integration tests across the Spring stack, identified a missing route that incorrectly returned 405 instead of 404.

Querybird - Natural Language to SQL | Python, FastAPI, SQLite, LLM (Anthropic), JavaScript July 2026

- Engineered a natural-language-to-SQL system using an LLM with schema-grounded prompting, reaching 64.2% execution accuracy on the full 500-question BIRD Mini-Dev benchmark, scored with the official evaluator.
- Developed a failure-analysis harness that categorized all 500 model outputs, finding ~99% of failures were semantic rather than execution errors and prioritizing prompt engineering over error-retry strategies.
- Deployed it as a full-stack web app (FastAPI + vanilla JS) with model-generated SQL sandboxed behind read-only, SELECT-only execution with query timeouts and row caps.

Macro Meal Optimizer | Python, FastAPI, PostgreSQL, React, PuLP, Docker June 2026

- Developed a meal optimization system that achieved calorie and macro targets within 0.3% using linear programming over a 5,200-item USDA food database, with a median solve time of 184ms.
- Optimized meal recommendations by resolving equally optimal solutions with weighted food preferences, reducing a typical meal from 11 garnish items to 3 practical ingredients while maintaining macro accuracy.
- Implemented 82 tests covering solver correctness, API integration, and UI behavior using pytest and Vitest, achieving 98% statement coverage across backend modules.

Premier League Predictor | Python, FastAPI, scikit-learn, Render January - May 2026

- Shipped a full-stack Premier League prediction app (FastAPI + vanilla JS, deployed on Render) that generates on-demand win/draw/loss probabilities, expected goals, and scoreline distributions for any fixture.
- Achieved 53.9% 1X2 accuracy and 0.581 Brier score across a 1,711-match walk-forward backtest using a combined Elo + Poisson model, beating a home-win baseline by ~9 percentage points.
- Modelled expected goals and scoreline probabilities per fixture using Poisson regression over team attack/defence rates, combined with Elo ratings updated match-by-match across five seasons of data (2021–2026).

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, TypeScript, R
Frameworks & Libraries: FastAPI, React, Next.js, Spring Boot, Pydantic, scikit-learn, Pandas, NumPy, PuLP
ML & AI: Machine Learning, Feature Engineering, Model Evaluation, NMF, Poisson Regression, LLMs, Prompt Engineering
Backend & Databases: REST APIs, PostgreSQL, SQLite, JPA/Hibernate
Tools and Testing: Git, Docker, pytest, Vitest, JUnit, Maven, Render

CERTIFICATIONS

IBM AI Engineering Professional Certificate | AWS Cloud Practitioner Essentials | Microsoft Power BI